



PRODUCT DATA SHEET

REFRIGERATION OIL – TECHNOFLUID REFRIGERATION OIL (Grades 46 & 68)

Commercial / Pack Name: REFRIGERATION OIL 46 / 68

Product Description

- TECHNOFLUID Refrigeration Oils 46 & 68 are premium-quality refrigeration compressor oils formulated from highly refined base oils with advanced additive technology.
- Designed to provide excellent lubricity, chemical stability, and thermal performance under low-temperature operating conditions.
- Exhibit high resistance to oxidation, ensuring long oil life and reliable compressor performance.
- Offer excellent miscibility and compatibility with commonly used refrigerants.
- Maintain fluidity at low temperatures, ensuring smooth compressor operation and reduced wear.

Key Performance Benefits

- Excellent lubricating film strength for effective wear protection of compressor components.
 - High chemical and thermal stability minimizes sludge, carbon, and deposit formation.
 - Low pour point ensures reliable lubrication during low-temperature start-up.
 - Superior oxidation resistance extends oil drain intervals and reduces maintenance.
 - Helps maintain system efficiency by keeping heat transfer surfaces clean.
 - Provides reliable sealing and lubrication of moving parts under continuous operation.
-

Applications

- Recommended for refrigeration and air-conditioning compressors.
- Suitable for use in:
 - Cold storage plants
 - Ice plants and chillers
 - Commercial refrigeration systems
 - Industrial refrigeration units
 - HVAC systems requiring ISO VG 46 or ISO VG 68 grades

- Compatible with reciprocating, rotary, and screw compressors (as per OEM recommendation).

Operating Characteristics

- Excellent low-temperature flow properties for uninterrupted lubrication.
- High dielectric strength (where applicable) supports safe electrical operation in sealed systems.
- Compatible with commonly used system materials, seals, and metals.

Typical Characteristics

Characteristics	Test Method	ISO VG 46	ISO VG 68
Appearance	Visual	Clear & Bright	Clear & Bright
Colour	ASTM D1500	≤ 0.5	≤ 0.5
Base Oil Type	—	Highly Refined Mineral	Highly Refined Mineral
Density @ 15°C, g/ml	ASTM D4052	0.851	0.858
Kinematic Viscosity @ 40°C, cSt	ASTM D445	46	68
Kinematic Viscosity @ 100°C, cSt	ASTM D445	6.6	8.7
Viscosity Index	ASTM D2270	≥ 95	≥ 95
Pour Point, °C	ASTM D97	-30	-24
Flash Point (COC), °C	ASTM D92	≥ 210	≥ 220
Neutralization Value, mg KOH/g	ASTM D664	≤ 0.03	≤ 0.03
Moisture Content, ppm	ASTM D6304	≤ 50	≤ 50
Copper Corrosion (3 hrs @ 100°C)	ASTM D130	1a	1a
Sulphur Content	ASTM D4294	Nil	Nil
Ash Content, % wt	ASTM D482	≤ 0.01	≤ 0.01
Oxidation Stability	ASTM D2440	Excellent	Excellent

Refrigerant
Compatibility

—

CFC / HCFC

CFC / HCFC